

Cyber Security in Banking Sector

1. Introduction to Cyber Security

Definition of Cyber Security

Cyber Security is the practice of protecting computers, networks, applications, devices, and digital data from unauthorized access, cyber attacks, damage, theft, and misuse. In simple words, Cyber Security protects digital information and computer systems from cyber threats. With the increasing use of computers, smartphones, internet services, online banking, cloud storage, and digital communication, protecting digital assets has become extremely important.

Cyber Security ensures that:

- Data remains safe
- Systems work properly
- Unauthorized people cannot access information
- Users can safely communicate and perform online activities

Why Cyber Security is Needed

Today, individuals and organizations store large amounts of sensitive information digitally, including:

- Personal details
- Banking information
- Business records
- Medical information
- Government data
- Confidential documents

Cyber criminals attempt to steal, damage, or misuse this information through different types of cyber attacks.

Cyber Security provides protection against:

- Data theft
- Identity theft
- Malware attacks
- Online fraud
- Unauthorized access
- Information leakage

2. Importance of Cyber Security

Cyber Security is important because it protects individuals, businesses, and governments from cyber threats.

1. Protection of Personal Information

Cyber Security protects personal information such as:

- Name
- Address
- Passwords
- Bank details
- Identification documents

Without proper security, attackers can steal personal information and misuse it for fraud or identity theft.

2. Protection of Business Data

Organizations store important information such as:

- Customer details
- Financial records
- Employee information
- Business strategies

Cyber Security prevents unauthorized access and protects business operations.

3. National Security Protection

Governments store sensitive information related to:

- Defence systems
- Public infrastructure
- Citizen records
- National projects

Cyber attacks on government systems can affect national security.

Cyber Security helps protect critical national infrastructure.

4. Prevention of Financial Losses

Cyber attacks can cause huge financial damage through:

- Online fraud
- Ransomware attacks
- Banking scams
- Data theft

Strong security measures help prevent financial losses.

5. Safe Digital Communication

Cyber Security ensures safe communication through:

- Email protection
- Secure messaging
- Encrypted communication

It prevents attackers from intercepting private conversations and sensitive information.

3. Basic Cyber Security Terminology

Understanding basic terms is important to understand cyber attacks and security.

1. Asset

Definition:

An **asset** is anything valuable that needs protection.

Assets may include:

- Data
- Computers
- Servers
- Applications
- Networks
- Digital accounts

Example:

A company's customer database is an important asset because it contains valuable information.

2. Threat

Definition:

A **threat** is anything that can cause damage to a system, network, or data.

A threat may be:

- A hacker
- Malware
- Data theft attempt
- Natural disaster
- Insider misuse

3. Vulnerability

Definition:

A **vulnerability** is a weakness or security flaw that attackers can exploit.

Common vulnerabilities include:

- Weak passwords
- Outdated software
- Poor security settings
- Unprotected networks

4. Risk

Definition:

Risk is the possibility of loss or damage when a threat takes advantage of a vulnerability.

Risk depends on:

- Presence of threat
- Existing vulnerability
- Possible impact

5. Exploit

Definition:

An **exploit** is a method or technique used by attackers to take advantage of a vulnerability.

Attackers use exploits to:

- Enter systems
- Install malware
- Steal information

Relationship Between Cyber Security Terms



4. CIA Triad in Cyber Security

The **CIA Triad** is the foundation of information security.

CIA stands for:

- Confidentiality
- Integrity
- Availability

It represents the three main goals of Cyber Security.

1. Confidentiality

Definition:

Confidentiality ensures that information is available only to authorized users.

It prevents unauthorized people from accessing sensitive data.

Examples:

- Password protection
- Encryption
- Access control

Importance of Confidentiality

It protects:

- Personal information
- Business secrets
- Financial data
- Government information

2. Integrity

Definition:

Integrity ensures that information remains accurate, complete, and unchanged.

It protects data from:

- Unauthorized modification
- Accidental changes
- Data corruption

Examples:

- Digital signatures
- Hashing
- Data verification

Importance of Integrity

Maintains:

- Accuracy of information
- Trustworthiness of data
- Reliability of systems

3. Availability

Availability ensures that systems and data are accessible whenever required.

A secure system should allow authorized users to access information without unnecessary interruption.

Examples:

- Backup systems
- Disaster recovery plans
- Regular maintenance

Example:

A banking website should remain available for customers to perform transactions.



CIA Triad Summary Table

Principle	Meaning	Purpose	Example
Confidentiality	Protect information from unauthorized access	Keep data private	Password protection
Integrity	Maintain accuracy of information	Prevent unauthorized changes	Digital signatures
Availability	Ensure access when needed	Keep systems running	Backup systems

B. TYPES OF CYBER SECURITY, CLOUD STORAGE & CYBER THREATS

5. Types of Cyber Security

Cyber Security is divided into different categories based on what needs to be protected.

The major types of Cyber Security are:

1. Network Security
2. Application Security
3. Information Security
4. Cloud Security
5. Mobile Security
6. IoT Security

1. Network Security

Definition:

Network Security protects computer networks from unauthorized access, hackers, cyber attacks, and misuse. A network connects multiple computers and devices to share information. Attackers often try to enter networks to steal data or damage systems.

Network Security uses different techniques to protect networks.

Common Network Security Methods

1. Firewall

A firewall is a security system that monitors and controls incoming and outgoing network traffic.

It acts as a barrier between a trusted network and an untrusted network.

Example: A firewall blocks unauthorized users from accessing a company's internal network.

2. Secure Wi-Fi Networks

Wi-Fi networks should be protected using:

- Strong passwords
- WPA/WPA2/WPA3 security protocols
- Regular password changes

3. Network Monitoring

Organizations continuously monitor network activity to identify suspicious behaviour.

Example of Network Security

Firewalls and secure Wi-Fi systems protect networks from unauthorized access.

2. Application Security

Definition:

Application Security protects software applications from:

- Bugs
- Malware
- Unauthorized access
- Security vulnerabilities

Applications may contain weaknesses that hackers can exploit.

Application Security ensures software remains safe and reliable.

Importance of Application Security

It protects:

- User information
- Application functionality
- Business data

Application Security Methods

1. Secure Coding

Developers write software using secure programming practices to reduce vulnerabilities.

2. Software Updates

Regular updates fix security weaknesses and improve protection.

3. Secure Login Systems

Applications use:

- Password authentication
- Multi-factor authentication
- Access permissions

Example:

Secure login systems and regular software updates protect applications.

3. Information Security

Definition:

Information Security protects important data from:

- Theft
- Loss
- Unauthorized access
- Unauthorized modification

Information security focuses on maintaining:

- Confidentiality
- Integrity
- Availability

Types of Information Protected

Examples include:

- Personal information
- Financial records
- Business documents
- Medical records

Methods Used in Information Security

1. Encryption

Encryption converts readable information into a protected form that unauthorized people cannot understand. Only authorized users with the correct key can access the original information.

2. Access Control

Access control ensures that only authorized people can access specific information. **Example:** An employee may access only the files required for their job.

Example: Encryption and access control protect confidential information.

4. Cloud Security

Cloud Security protects data and services stored on cloud platforms from cyber threats. Cloud services allow users to store and access data through the internet. As cloud storage contains important information, it requires strong security measures.

Cloud storage is a service that allows users to store files and data on the internet instead of saving them only on their computer or mobile phone.

The stored data is kept on:

- Remote servers
- Powerful computers
- Data centres managed by companies

Users can access their files anytime using the internet.

Examples of Cloud Storage

1. Google Drive

A cloud storage service provided by Google that allows users to store:

- Documents
- Photos
- Videos
- Files

2. OneDrive

A cloud storage service provided by Microsoft.

It allows users to save and synchronize files across devices.

3. iCloud

A cloud storage service provided by Apple.

It stores:

- Photos
- Contacts
- Documents
- Device backups

4. Dropbox

A cloud storage platform used for storing and sharing files online.



How Cloud Storage Works

Step 1: Upload Data

The user uploads a file such as:

- Photo
- Document
- Video

to the cloud.



Step 2: Storage on Remote Servers

The file is stored on secure internet servers managed by cloud providers.



Step 3: Access Anytime

The user can access the file from any device using their account and internet connection.

Advantages of Cloud Storage

1. Access Files Anywhere

Users can access their files from:

- Laptop
- Smartphone
- Tablet

from any location.

2. Backup Data Safely

Cloud storage provides backup protection against:

- Device damage
- Data loss
- Accidental deletion

3. Easy Sharing

Users can quickly share files with others through links or permissions.

4. Saves Storage Space

Files stored online reduce the need for large storage capacity on personal devices.

5. Mobile Security

Definition:

Mobile Security protects smartphones and tablets from:

- Viruses
- Data theft
- Unauthorized access
- Malicious applications

Mobile devices contain sensitive information such as:

- Photos
- Contacts
- Banking applications
- Personal messages

Therefore, mobile security is essential.

Mobile Security Methods

1. Screen Lock

Using:

- Password
- PIN
- Fingerprint
- Face recognition

prevents unauthorized access.

2. App Permissions

Users should carefully check permissions requested by applications.

Example: A flashlight application should not require access to contacts or messages.

3. Security Updates

Regular updates fix vulnerabilities and improve device protection.

6. IoT Security (Internet of Things Security)

Definition:

IoT Security protects smart devices connected to the internet from hacking and misuse.

IoT devices collect and exchange data through networks.

Examples of IoT Devices

- Smart cameras
- Smart TVs
- Smart watches
- Smart home systems
- Smart speakers

Risks in IoT Devices

IoT devices may be attacked because of:

- Weak passwords
- Outdated software
- Poor security settings

IoT Security Methods

1. Strong Passwords

Avoid default passwords and create unique passwords.

2. Regular Updates

Install security updates provided by manufacturers.

3. Secure Network Connection

Connect IoT devices only to trusted networks.

Example:

Smart cameras and smart home devices require secure passwords and updates to prevent hacking.

Cyber Security in Banking Sector Part 2

Cyber Threats

A **cyber threat** is any possible danger that can damage computer systems, networks, applications, or digital information.

Cyber threats are created by attackers to:

- Steal information
- Damage systems
- Gain unauthorized access
- Cause financial losses

Major Types of Cyber Threats

The major cyber threats include:

1. Malware
2. Phishing
3. Password Attacks
4. DoS and DDoS Attacks
5. Data Breaches
6. Social Engineering

1. Malware

Definition:

Malware means Malicious Software.

It refers to harmful programs created to:

- Damage devices
- Steal information
- Control systems without permission

Examples of Malware

- Virus
- Worm
- Trojan
- Ransomware
- Spyware

2. Phishing

Definition:

Phishing is a cyber attack where attackers create fake:

- Emails
- Messages
- Websites

to trick users into sharing sensitive information.

Information targeted:

- Passwords
- Bank details
- OTPs
- Personal information

Example:

A fake bank email:

"Your account will be blocked. Click here to verify."

The link may lead to a fake website designed to steal login details.

3. Password Attacks

Password attacks are attempts by hackers to:

- Guess passwords
- Steal passwords
- Break account security

4. DoS and DDoS Attacks

Denial of Service (DoS)

A DoS attack sends excessive requests to a server to make a service unavailable.

Distributed Denial of Service (DDoS)

A DDoS attack uses many infected devices to attack a target system.

5. Data Breaches

A data breach occurs when unauthorized people access or steal private information.

Examples of Stolen Data

- Customer names
- Passwords
- Financial details
- Personal records

6. Social Engineering

Definition:

Social Engineering is a technique where attackers manipulate people instead of directly attacking computer systems.

Attackers use:

- Trust
- Fear
- Urgency
- False identity

to trick victims.

Example:

A person pretending to be a bank employee and asking for an OTP.

PART 3: MALWARE, SOCIAL ENGINEERING, PASSWORD ATTACKS & DATA SECURITY PROTECTION

7. Malware

Malware stands for **Malicious Software**.

Malware refers to harmful software programs created by cyber criminals to:

- Damage computer systems
- Steal sensitive information
- Monitor user activities
- Gain unauthorized access
- Control devices without permission

Malware can affect:

- Computers
- Smartphones
- Servers
- Networks
- IoT devices

How Malware Enters a System

Malware can enter devices through:

- Malicious email attachments
- Fake websites
- Infected software downloads
- Unsafe links
- External storage devices

- Pirated applications

Types of Malware

The major types of malware are:

1. Virus
2. Worm
3. Trojan
4. Ransomware
5. Spyware

1. Virus

Definition:

A **virus** is a type of malware that attaches itself to files or programs and spreads when the infected file is opened or shared. A virus usually requires human action to spread.

Virus attaches to a legitimate file or program.

↓

User opens the infected file.

↓

Virus becomes active.

↓

It damages files or spreads to other systems.

Effects of Virus:

A virus can:

- Delete files
- Corrupt data
- Slow down computers
- Damage operating systems

Example:

Opening an infected document attachment that installs harmful software.

2. Worm

A **worm** is malware that can copy itself and spread automatically through computer networks without requiring human action.

Unlike viruses, worms do not need to attach themselves to files.

How Worm Works:

Finds a vulnerable system.



Exploits the weakness.



Copies itself automatically.



Spreads to other connected devices.

3. Trojan

A **Trojan** is malware that appears to be a useful or legitimate software but secretly performs harmful activities. It gets its name from the Trojan Horse story because it hides malicious activity inside something that looks safe.

How Trojan Works:

User downloads a fake application.



The program appears useful.



Once installed, it performs harmful actions.

Effects of Trojan:

- Steals passwords
- Installs other malware
- Gives attackers remote access
- Damages files

4. Ransomware

Ransomware is malware that locks or encrypts a user's files and demands money (ransom) to restore access.

How Ransomware Works:

Malware enters the system.

↓

Important files are encrypted.

↓

User loses access to files.

↓

Attacker demands payment.

Effects of Ransomware:

- Loss of important data
- Business interruption
- Financial losses

5. Spyware

Spyware is malware that secretly monitors user activities and collects information without permission.

Information Collected by Spyware:

- Passwords
- Browsing history
- Personal details
- Banking information

Effects of Spyware:

- Privacy loss
- Identity theft
- Financial fraud

8. Social Engineering Attacks

Social Engineering attacks are cyber attacks where criminals target people instead of directly attacking computers or systems.

Attackers use:

- Tricks
- Lies
- Manipulation
- False identities

to make people reveal confidential information or perform unsafe actions.

Why Social Engineering Works

Attackers exploit human emotions such as:

- Fear
- Trust
- Curiosity

- Urgency
- Greed

Examples of Information Targeted:

- Passwords
- OTPs
- Bank details
- Personal information
- Company secrets

Types of Social Engineering Attacks

Major types include:

1. Phishing
2. Spear Phishing
3. Smishing
4. Vishing
5. Baiting

1. Phishing

Phishing is a cyber attack where attackers send fake emails, messages, or create fake websites to trick people into sharing sensitive information.

Common Targets:

- Passwords
- Bank details
- OTPs
- Login credentials



How Phishing Works:

Attacker creates a fake message.



Victim receives the message.



Victim clicks a malicious link.



Information is stolen.

Example:

"You have won a prize! Click this link and enter your bank details."

2. Spear Phishing

Spear phishing is a targeted phishing attack where attackers create personalized messages for a specific person or organization. Unlike normal phishing, it does not target random people.

Features:

- Highly personalized
- Targets specific individuals
- Uses collected information about the victim

3. Smishing

Definition:

Smishing means SMS + Phishing.

It is a phishing attack performed through text messages.

How Smishing Works:

Attackers send fake SMS messages containing:

- Malicious links
- Fake offers
- Urgent warnings

Example:

"Your bank account will be blocked. Click this link to update your KYC details."

4. Vishing

Definition:

Vishing means Voice + Phishing.

It is a phishing attack performed through phone calls.

Attackers pretend to be trusted individuals.

Common Pretenders:

- Bank employees
- Government officials
- Technical support staff

Example:

A caller pretending to be a bank employee asks for OTP or PIN details.

5. Baiting

Definition:

Baiting is an attack where criminals offer something attractive to trick people into performing unsafe actions.

The attacker uses a "bait" to attract victims.

Protection Against Social Engineering Attacks

Users should:

- Verify suspicious messages
- Never share OTPs or passwords
- Avoid clicking unknown links
- Confirm requests from trusted people
- Be careful with unexpected offers

9. Password Attacks

Definition:

Password attacks are attempts by hackers to gain unauthorized access by:

- Guessing passwords
- Stealing passwords
- Breaking password protection

Common Password Attacks

1. Brute Force Attack
2. Dictionary Attack
3. Credential Stuffing

1. Brute Force Attack

Definition:

A brute force attack attempts to guess a password by trying every possible combination until the correct password is found.

Example:

Trying millions of password combinations automatically.

Protection:

- Use long passwords
- Enable account lock after multiple failures
- Use multi-factor authentication

2. Dictionary Attack

Definition:

A dictionary attack uses a list of commonly used passwords to guess the correct password.A hacker tries a list of common passwords to access an account.

3. Credential Stuffing

Definition:

Credential stuffing uses stolen usernames and passwords from one website to access accounts on another website.

Why It Works:

Many users reuse the same password on multiple websites.

Example:

A stolen Facebook password is used to access a Gmail account.

Password Attack Comparison Table

Attack	How It Works	Example
Brute Force	Tries every possible combination	Millions of password attempts
Dictionary Attack	Uses common password lists	Trying "password123"
Credential Stuffing	Uses leaked passwords	Using stolen Facebook password on Gmail

10. Data Security Protection Methods

Definition of Data Security

Data security means protecting personal and important information from:

- Theft
- Loss
- Unauthorized access
- Misuse

1. Use Strong Passwords

Strong passwords are difficult for attackers to guess.

A strong password should contain:

- ✓ Capital letters
- ✓ Small letters
- ✓ Numbers
- ✓ Symbols

2. Enable Two-Factor Authentication (2FA)

Definition:

Two-factor authentication adds an additional security layer after entering a password.

Working:

Step 1:

Enter username and password.

↓

Step 2:

Enter OTP or verification code.

Example:

Password + OTP sent to mobile phone.

3. Do Not Share Sensitive Information

Never share:

- Passwords
- OTPs
- PIN numbers
- Banking details

Remember:

Banks never ask customers for OTPs.

4. Keep Software Updated

Software updates:

- Fix security weaknesses
- Remove vulnerabilities
- Improve protection

5. Use Antivirus and Security Tools

Antivirus software helps:

- Detect malware
- Remove viruses
- Protect devices

6. Be Careful with Emails and Links

Avoid:

- Unknown attachments
- Suspicious links
- Fake websites

because they may contain phishing attacks or malware.

7. Backup Important Data

Backup means keeping additional copies of important files.

Backups help recover data after:

- Device failure
- Malware attacks
- Accidental deletion

Examples:

- External hard drive backup
- Cloud storage backup

8. Secure Wi-Fi Networks

Protect Wi-Fi by:

- Using strong passwords
- Changing default router passwords
- Avoiding unknown users

9. Encrypt Sensitive Data

Definition:

Encryption converts readable data into a protected form that cannot be understood without permission.

Example:

Online banking uses encryption to protect transactions.

10. Lock and Protect Devices

Use:

- Screen locks
- Fingerprints
- Face recognition
- PIN protection

to prevent unauthorized access.